Summary in Graph

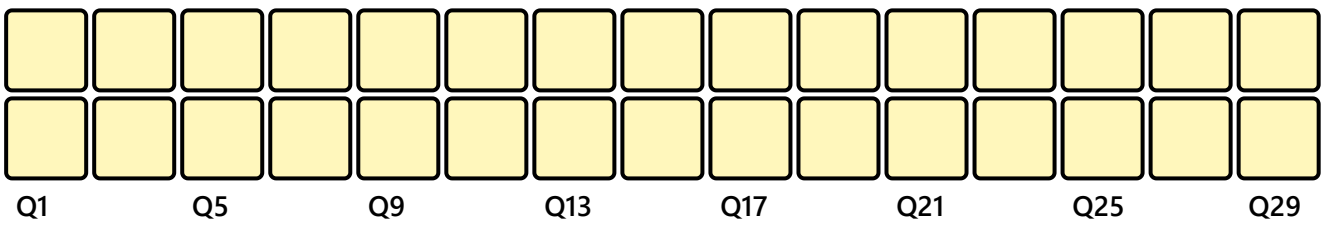
# Exam Summary (GO Classes GATE CS/DA | Calculus | Weekly Quiz 1)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 30 (15 + 15) |
| Total Marks: | 45 (15 + 30) |
| Exam Duration: | 120 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (39) out of 391 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Let $f$ be a continuous function defined for all real numbers. Suppose

$$\int_0^3 f(x)dx = 2, \quad \int_1^2 f(x)dx = 7, \quad \int_3^6 f(x)dx = -5, \quad \text{and} \quad \int_9^{18} f(x)dx = -3$$

Find $\int_3^6 f\left(\frac{1}{3}t\right) dt$

Your Answer: | Correct Answer: 21 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Which of the following represents the value of $\int_1^e \frac{\ln(x)}{x} dx$?

A. $1$
B. $2$
C. $1/2$
D. $e$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Find $\int xe^{-3x} dx$

A. $\frac{1}{2}e^{-3x}(x-3)+C$

B. $-\frac{1}{9}e^{-3x}(3x+1)+C$

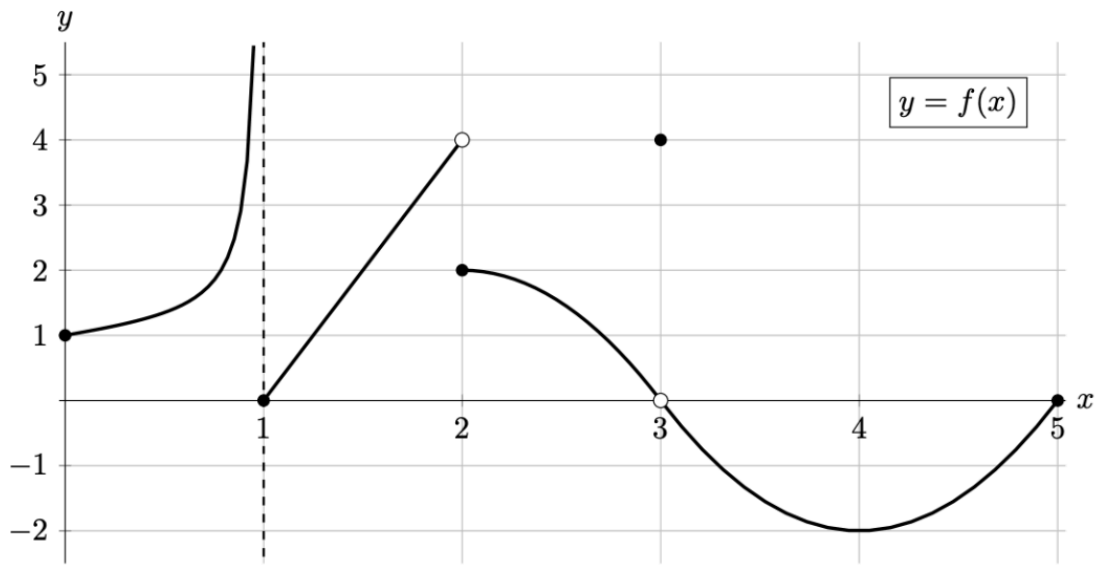
C. $-\frac{1}{6}x^2e^{-3x}+C$

D. $-\frac{1}{3}e^{-3x}(x+1)+C$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Below is a portion of the graph of an even function $f(x)$, which has domain $(-\infty, \infty)$ even though the graph below only shows the function on the interval $[0, 5]$. Note that $f(x)$ has a vertical asymptote at $x = 1$.




Find

$$\lim_{h \to 0} \frac{f(1.5+h) - f(1.5)}{h}$$

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #5** Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Some information about the derivative $p'(x)$ and the second derivative $p''(x)$ of a function $p(x)$ is provided in the table below.

| $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|---|---|
| $p'(x)$ | $1$ | $0$ | $-2$ | $0$ | $-1$ | $0$ | $2$ |
| $p''(x)$ | $-1$ | $0$ | $0$ | $0$ | $0$ | $2$ | $1$ |

At which of the following values of $x$ must $p(x)$ have a local minimum?

A. $x = -3$
B. $x = -2$
C. $x = -1$
D. $x = 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #6** Numerical Type | Award: 1 | Penalty: 0 | Calculus

Consider the piecewise function

$$q(x) = \begin{cases} 7e^{x-C} + \dfrac{x}{3x-2} & x < 0 \\ \dfrac{6+5x}{2+3x+4^x} & x \geq 0 \end{cases}$$

where $C$ is a constant. Find $\lim_{x \to \infty} q(x)$

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #7** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function $g(x)$ is given by the equation

$$g(x) = \begin{cases} ax^2 & x \leq 1 \\ b - \ln(3x) & x > 1 \end{cases}$$

where $a$ and $b$ are constants.
Find the value of $a$ such that function is differentiable at $x = 1$.

A. $1/2$
B. $-1/2$
C. $2$
D. $-2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #8** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the family of functions

$$g(x) = a\ln(x) + \frac{b}{x}$$

defined for $x > 0$, where $a$ and $b$ are positive constants.

Any function $g(x)$ in this family has only one critical point. In terms of $a$ and $b$, what is the $x$-coordinate of that critical point?

A. $x = b/a$ is a critical point where the function shows local minima
B. $x = b/a$ is a critical point where the function shows local maxima
C. $x = a/b$ is a critical point where the function shows local minima
D. $x = a/b$ is a critical point where the function shows local maxima

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #9** Numerical Type | Award: 1 | Penalty: 0 | Calculus

The following are tables of values for two differentiable functions $f(x)$ and $g(x)$ and their derivatives. Missing values are denoted by a "?". Assume that each of these functions is defined for all real numbers, that $f'(x)$ and $g'(x)$ are continuous.

| $x$ | 0 | 2 | 3 | 6 | 9 |
|---|---|---|---|---|---|
| $f(x)$ | $-1$ | ? | 0 | $-2$ | ? |
| $f'(x)$ | 1 | 4 | $-1$ | ? | 1 |

| $x$ | $-1$ | 1 | 3 | 7 | 11 |
|---|---|---|---|---|---|
| $g(x)$ | $-4$ | 1 | 2 | 6 | 7 |
| $g'(x)$ | 7 | ? | 3 | 4 | ? |

Let $z(x) = f(g(x))$. Find $z'(3)$

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #10** Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Which of the following is/are TRUE?

A. If $\lim_{x\to 5} f(x) = 0$ and $\lim_{x\to 5} g(x) = 0$, then $\lim_{x\to 5} \frac{f(x)}{g(x)}$ does not exist.

B. If $f, g$, are any two functions which are continuous for all $x$, then $\frac{f}{g}$ is continuous for all $x$.

C. It is possible that functions $f$ and $g$ are not continuous at a point $x_0$, but $f+g$ is continuous at $x_0$.

D. If $f'(c) = 0$ then $f(x)$ has a local maximum or a local minimum at $x = c$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #11** Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Suppose $f$ is a function such that $f'(x) = 4x^3$ and $f''(x) = 12x^2$.

Which of the following is /are true?

A. $f$ has a local maximum at $x = 0$ by the first derivative test

B. $f$ has a local minimum at $x = 0$ by the first derivative test

C. $f$ has a local maximum at $x = 0$ by the second derivative test

D. $f$ has a local minimum at $x = 0$ by the second derivative test

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #12** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The derivative of $g(x) = e^{\sqrt{x}}$ is

A. $\sqrt{x}e^{\sqrt{x}-1}$

B. $2e^{\sqrt{x}}x^{-0.5}$

C. $\frac{0.5e^{\sqrt{x}}}{\sqrt{x}}$

D. $e^{\sqrt{x}}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #13** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function

$$f(x) = \begin{cases} e^x & \text{if} \quad x \leq 1 \\ mx + b & \text{if} \quad x > 1 \end{cases}$$

is continuous and differentiable at $x = 1$.

Find the value of $m - b$?

A. $e$

B. $-e$

C. $\mathrm{e} - 1$

D. $1 - e$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #14** Numerical Type | Award: 1 | Penalty: 0 | Calculus

Given $F(x) = (f(g(x)))^2, g(1) = 2, g'(1) = 3, f(2) = 4$, and $f'(2) = 5$, find $F'(1)$

Your Answer: | Correct Answer: 120 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #15** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Find the values of $A$ and $B$ that make

$$f(x) = \begin{cases} x^2 + 1 & \text{if} \quad x \geq 0 \\ A \sin x + B \cos x & \text{if} \quad x < 0 \end{cases}$$

differentiable at $x = 0$.

A. $A = 0, B = 1$
B. $A = 1, B = 0$
C. $A = 0, B = -1$
D. $A = -1, B = 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #16** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Is there a number $b$ such that $\lim_{x \to -2} \frac{bx^2 + 15x + 15 + b}{x^2 + x - 2}$ exists? If so, find the value of the limit.

A. $-1$
B. $-2$
C. $1$
D. There is no such $b$ for that above limit exist

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #17** Numerical Type | Award: 2 | Penalty: 0 | Calculus

$$\lim_{x \to \infty} (x + \sin x)^{\frac{1}{x}}$$

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #18** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $g$ is continuous (but not differentiable) at $x = 0, g(0) = 8$, and $f(x) = xg(x)$, find $f'(0)$.

A. $0$
B. $8$

C. $1$
D. $f(x)$ is also not differentiable at $x = 0$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

**Q #19** Numerical Type | Award: 2 | Penalty: 0 | Calculus

Suppose that $f(x)$ and $g(x)$ are differentiable functions and that $h(x) = f(x)g(x)$. You are given the following table of values:

| | |
|---|---|
| $h(1)$ | $24$ |
| $g(1)$ | $6$ |
| $f'(1)$ | $-2$ |
| $h'(1)$ | $20$ |

Using the table, find $g'(1)$.

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #20** Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Suppose $f(x)$ is continuous and diffrentiable for all $x$. And $-1 \leq f'(x) \leq 3$ fo all $x$. Which of the following is/are ALWAYS true?

A. $f(5) \leq f(3) + 6$
B. $f(5) \geq f(3) - 2$
C. $f(5) \leq f(3) + 10$
D. $f(5) \geq f(3) - 10$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #21** Numerical Type | Award: 2 | Penalty: 0 | Calculus

Define

$$g(x) = \begin{cases} -6x + 2 & x \leq 2 \\ x - 2 & x > 2 \end{cases}$$

Find $\int_1^4 g(x)dx$.

Your Answer: | Correct Answer: -5 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #22** Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following is/are TRUE?

A. $\int_{-5}^{5} \left(ax^2 + bx + c\right) dx = 2\int_{0}^{5} \left(ax^2 + c\right) dx$

B. If $f$ and $g$ are continuous and $f(x) \geqslant g(x)$ for $a \leqslant x \leqslant b$, then

$$\int_a^b f(x)dx \geqslant \int_a^b g(x)dx$$

C. If $f$ and $g$ are differentiable, then

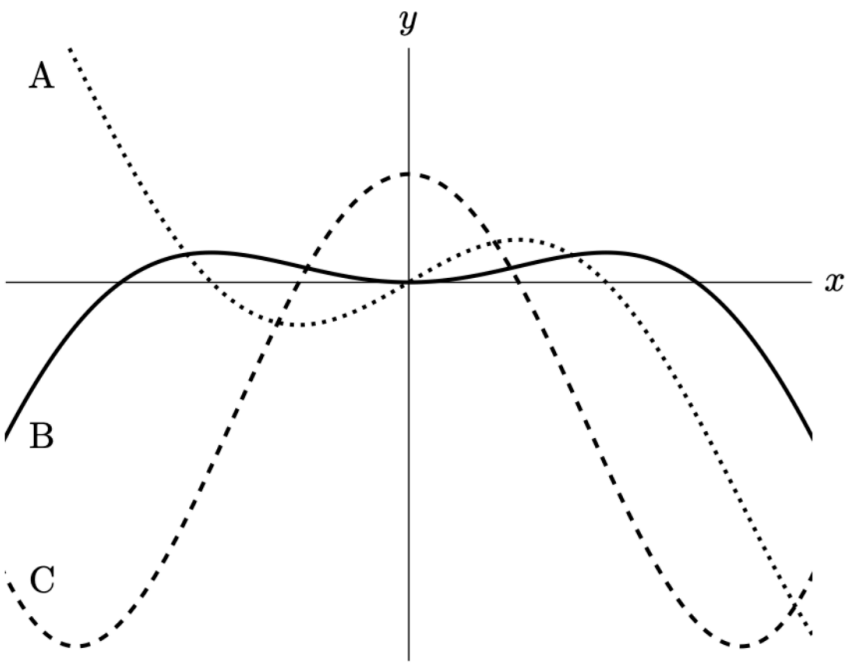
$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

D. If $f$ is differentiable, then $\frac{d}{dx}\sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}$.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #23** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Shown below are portions of the graphs of $y = f(x), y = f'(x)$, and $y = f''(x)$. Determine which graph is which.




A. $A - f(x), B - f'(x), C - f''(x)$
B. $A - f'(x), B - f(x), C - f''(x)$
C. $A - f''(x), B - f(x), C - f'(x)$
D. $A - f'(x), B - f''(x), C - f(x)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #24** Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

The function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} \frac{x}{x^2+1} & x \leq 0 \\ ? & x > 0 \end{cases}$$

Note that the formula for $f(x)$ for $x > 0$ is unknown.

However, it is known that $f(x)$ is differentiable at each point in its domain $(-\infty, \infty)$, and that $f'(x) > 0$ for all $x \geq 0$.

Which of the following option is/are TRUE?

A. $x = 0$ has global minima
B. $x = 0$ has global maxima
C. $x = -1$ has global maxima
D. $x = -1$ has global minima

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Suppose $h(x)$ is a function such that $h(x)$ has exactly three critical point. Two of which are shown in the table below.
Assume that both $h(x)$ and $h'(x)$ are differentiable on $(-\infty, \infty)$

| $x$ | 0 | 3 | 5 | 7 |
|---|---|---|---|---|
| $h(x)$ | 2 | ? | 4 | 4 |
| $h'(x)$ | $-1$ | 0 | 0 | ? |

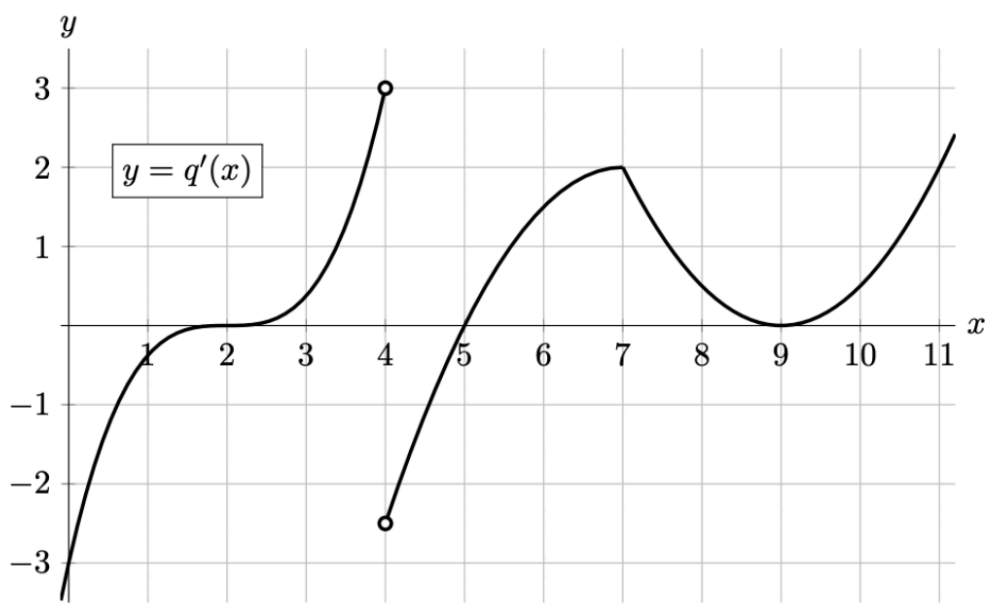
Further using Lagrange mean value theorem in the interval $[5, 7]$, we can determine the interval of the third critical point.
On which of the following intervals must $h(x)$ be increasing on the entire interval?

A. $(0, 3)$
B. $(3, 5)$
C. $(5, 6)$
D. $(6, 7)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let $q(x)$ be a continuous function which is defined for all real numbers. A portion of the graph of $q'(x)$, the derivative of $\boldsymbol{q(x)}$, is shown below.



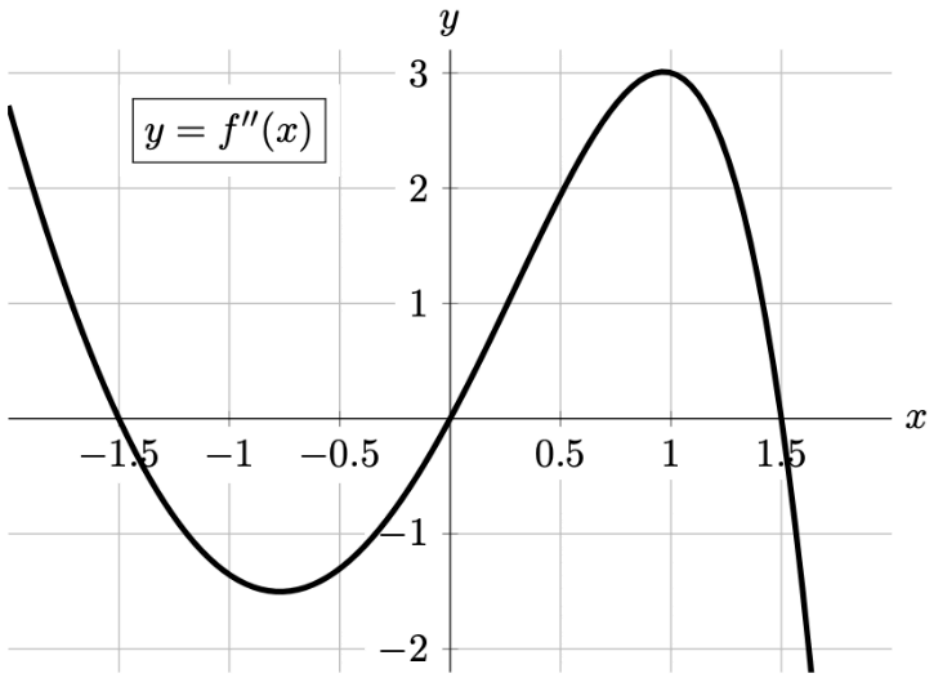

On which of the following interval(s) is $q''(x)$ positive?

A. $(0, 2)$
B. $(2, 4)$
C. $(7, 9)$
D. $(5, 7)$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Calculus

We consider a function $f(x)$ defined for all real numbers. We suppose that the first and second derivatives $f'(x)$ and $f''(x)$ are also defined for all real numbers. Below we show the graph of the second derivative of $f$. You may assume that $f''(x)$ is decreasing outside of the region shown.




Suppose that $f'(0) = 5$. How many critical points does $f$ have?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #28** Numerical Type | Award: 2 | Penalty: 0 | Calculus

Consider a continuous function $f(x)$, and suppose that $f(x)$ and its first derivative $f'(x)$ are differentiable everywhere. Suppose we know the following information about $f(x)$ and its first and second derivatives.

- On the interval $(-\infty, -2)$, we have $f(x) = 2^{-x}$.
- $\lim_{x\to\infty} f(x) = 6$
- $f(2) = -5, f(3) = 7$, and $f(4) = 8$
- $f'(x)$ is equal to 0 at $x = -1, 2, 4$, and not at any other $x$-values.
- $f''(x) < 0$ on the intervals $-1 < x < 0$ and $3 < x < 5$, and not on any other interval.

Find the global minimum of $f(x)$ on $(-\infty, \infty)$? (minimum value of $f(x)$)

Your Answer: | Correct Answer: -5 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #29** Numerical Type | Award: 2 | Penalty: 0 | Calculus

If $f(x) = (1+x)\left(1+x^2\right)\left(1+x^3\right)\left(1+x^4\right)$, then $f'(0) = ?$

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #30** Numerical Type | Award: 2 | Penalty: 0 | Calculus

Let $M = \lim_{x\to 0^+} \left(e^x + 3x\right)^{1/x}$. Find the value of $log_e M$.

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

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